The Ordinal Summit

Where the Tower Snaps to c_1 | Version 1.0

ZP Companion | May 2026

This companion explains the ideas in plain language with diagrams and examples. It is not the formal document — every claim here restates a result already proved in ZP-L Incomputability Convergence v1.0. Consult that document for the authoritative mathematics.

What Is ZP-L Doing?

ZP-K proved that the initial machine state c_0 is its own program — a Kleene fixed point, a self-executing Quine, the same object at the bottom of every formal language in the framework. ZP-L picks up the thread from a different direction: the ordinal direction. If a map ϕ assigns machine phases to ordinals — sending c_0 to everything below a certain point and c_1 above it — what is that threshold ordinal? ZP-L establishes the answer: ε_0, the first fixed point of the map $\alpha \mapsto \omega$^α.

This is not an arbitrary choice. ZP-L proves that ε_0 is the minimal threshold — no monotone, tower-aligned map can snap before ε_0, and the canonical map snaps exactly there. ZP-L also establishes the connection to ZP-B: as the tower stages approach ε_0 in ordinal order, their 2-adic encodings converge to $0 = \perp$ in $\mathbb{Z}_2$. The same sequence witnesses both convergences simultaneously.

Real-world analogy — the unreachable horizon

Imagine a path where each step is longer than the last: you walk 1 mile, then 2, then 4, then 8, ... You are always getting further from the start, but you are also always somewhere on the path — never at the horizon. The horizon (ε_0) is the limit of all those steps combined. It exists as a mathematical object, but no finite number of steps reaches it.

The ordinal tower works the same way. The snap that ZP-E proved occurs at ε_0 is not a step you can take from inside the tower. It is the point where the tower stops and something new begins.

What Is ε_0?

ε_0 (epsilon-zero) is the smallest ordinal α satisfying the equation ω^$\alpha = \alpha$. In other words, it is the first fixed point of the map that sends any ordinal α to ω^α (omega raised to the power α). This is not a coincidence or a definition by fiat — it is a property ε_0 satisfies and no smaller ordinal does.

The tower 0, 1, ω, ω^ω, ω^ω^ω, ... climbs through the ordinals, each stage being ω raised to the previous stage. Every stage is strictly below ε_0, and ε_0 is their limit — the supremum of the whole sequence. It is never reached from below; it is the boundary above all stages.

Each tower stage is strictly below ε-zero and maps to c0. The snap to c1 occurs exactly at ε-zero.

The tower stages 0, 1, ω, ω^ω, ... all lie strictly below ε_0 and map to c_0. The snap transition to c_1 occurs at the threshold ε_0 itself. The arrow marks the snap. ZP-L §VII: epsilon_zero_snap_canonical.

> Remember: ε_0 is not "infinity." It is a specific countable ordinal — the first one satisfying ω^α = α. Ordinals larger than ε_0 exist ($\varepsilon_0 + 1$, ε_1, and many more). ZP-L is not claiming that ε_0 is the largest ordinal or that the framework breaks down above it. It is the minimal snap threshold for monotone, tower-aligned maps — and that minimality is exactly what is proved.

Why ε_0 Is the Exact Threshold

Two facts together pin ε_0 as the minimal snap threshold. They work like a lower bound and an upper bound that meet at the same point.

Lower bound (snap cannot happen before ε_0): The fundamental sequence is cofinal in ε_0. This means: for any ordinal α below ε_0, some tower stage eventually exceeds α. A monotone map that assigns c_0 to every tower stage must therefore assign c_0 to α too (it cannot increase between a stage above α and α itself). So no ordinal strictly below ε_0 can be a snap point for such a map.

Upper bound (snap does happen at ε_0): ε_0 is itself a fixed point of ω^· (ω^ε_0 = ε_0). A map that assigns c_1 to all fixed points of ω^· must assign c_1 to ε_0. Together with the lower bound, ε_0 is the first place where c_1 can and must appear.

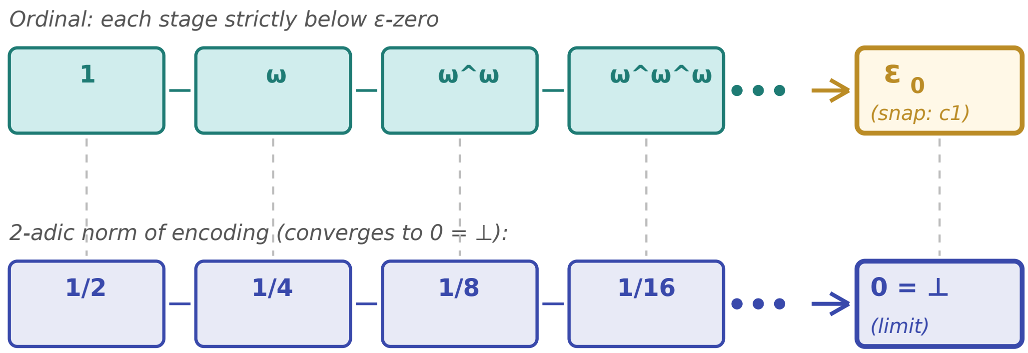
The canonical witness map $\phi(\alpha) = c_0$ if $\alpha < \varepsilon_0$, else c_1 satisfies all five conditions at once — monotonicity, tower-alignment, fixed-point-respecting, snapping at ε_0, and minimality — with no free hypotheses. This is the central result of ZP-L.

> "Minimal" does not mean "unique." ZP-L proves that no snap can occur strictly before ε_0 (under the stated conditions), not that ε_0 is the only possible snap point. Maps that also assign c_1 to ordinals above ε_0 are not ruled out. The framework fixes ε_0 as the threshold by the canonical witness, not by uniqueness of all satisfying maps.

The Two-Adic Connection

ZP-B established that $0 = \perp$ in the 2-adic integers $\mathbb{Z}_2$ is the valuative sink: the 2-adic valuation of 0 is infinite, which means 0 sits at the bottom of the 2-adic metric. ZP-L connects this to the ordinal tower via an encoding of ordinals below ε_0 into $\mathbb{Z}_2$ using their Cantor normal forms.

The encoding (cnfToZp2 in ZPL.lean) maps the n-th tower stage to 2^n in $\mathbb{Z}_2$. The 2-adic norm of 2^n is $(1/2)^n$, which goes to 0 as n increases. So as the ordinal tower climbs toward ε_0 from below, its 2-adic image descends toward $0 = \perp$ from above. Two convergences, one tower.



Same tower, two perspectives. Teal (top): ordinal stages approach ε-zero. Indigo (bottom): their 2-adic norms approach 0.

Same tower stages, two perspectives. Teal (top row): ordinal stages 1, ω, ω^ω, ... approach ε_0 from below. Indigo (bottom row): the 2-adic norms of their encodings 1/2, 1/4, 1/8, ... converge to $0 = \perp$. ZP-L §VII: snap_zp2_correspondence.

This dual convergence is what snap_zp2_correspondence formalizes in ZP-L §VII: four independently proved facts about the same tower sequence — all stages below ε_0 in ordinal order, all assigned c_0 by the canonical map, all encoding to 2^n in $\mathbb{Z}_2$ with 2-adic norm $(1/2)^n \to 0$, and the canonical map snapping to c_1 at ε_0.

The Kleene Connection

ZP-K identified ⊥ as a Kleene fixed point: a program that is its own output, guaranteed to exist by Kleene's second recursion theorem. ZP-L identifies ε_0 as an ordinal fixed point: the first ordinal satisfying ω^α = α, guaranteed to exist by the fixed-point theorem for normal functions. Both fixed-point proofs require Classical.choice at the same non-constructive step — the diagonal step where the fixed point is selected from an infinite collection.

ZP-L §VI (kleene_ordinal_snap_bridge) names this structural parallel explicitly. The theorem itself is purely ordinal — no Code or eval object appears in it. What the section establishes is that the hypothesis structure of the two proofs is the same: diagonalization, non-constructive selection, fixed-point existence.

Two rooms, same pattern

Imagine two mathematicians working in separate rooms. One (ZP-K) is computing with programs and asking: what program runs itself? The other (ZP-L) is computing with ordinals and asking: what ordinal satisfies ω^α = α? Neither knows what the other is doing. When they compare notes, they find the same proof structure: the same diagonal argument, the same non-constructive selection step, the same kind of fixed-point existence result.

ZPM (released with ZP-L) makes this precise by constructing a formal type bridge (snapEmbed) between the machine-phase and 2-adic settings, closing the gap that ZP-L's "Remaining Gap" section identified as outside Lean scope at time of writing. The two rooms share the same floor plan.

A Note on Classical.choice

Every ZP-L theorem carries the axiom footprint [propext, Classical.choice, Quot.sound]. This is the standard footprint of Lean 4 + Mathlib for any theorem that uses ordinal theory, p-adic analysis, or computability. It is not a novel ZP-L commitment.

What is notable is where Classical.choice appears. Across the four mathematical settings of the ZP framework — topology (ZP-B), information theory (ZP-C), set theory and computation (ZP-J/K), and ordinal theory (ZP-L) — the same axiom appears at the same structural step: the non-constructive diagonal. ZP-L §I documents this convergence. Whether Classical.choice is structurally forced by the ZP framework — rather than merely inherited from Mathlib infrastructure — remains an open question.

Key Result — ZP-L: The Ordinal Snap Threshold

epsilon_zero_snap_canonical (ZPL.lean §VII): The canonical map $\phi(\alpha) = c_0$ if $\alpha < \varepsilon_0$, else c_1 satisfies all five conditions simultaneously — monotone, tower-aligned (c_0 on every fundamental sequence stage), fixed-point-respecting (c_1 on every fixed point of ω^·), snapping at ε_0, and ε_0 minimal — with no free hypotheses. snap_zp2_correspondence co-proves that the same tower witnesses both the ordinal approach to ε_0 and the 2-adic convergence to $0 = \perp$. 24 theorems, zero sorry. ZPM closes the snapEmbed type bridge gap. ✓